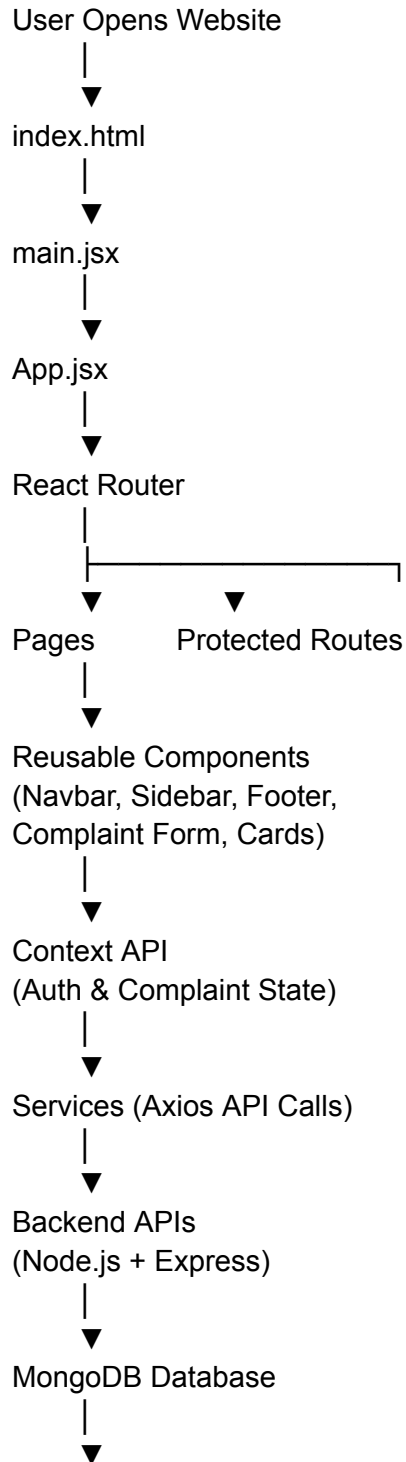
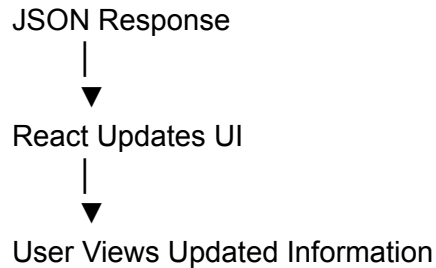


FRONTEND WORKFLOW





Frontend (Client) Development Explanation

Introduction

The **Client** folder contains the frontend of the **Online Complaint Registration System**, which is developed using **React.js**. It provides an interactive and user-friendly interface for **Users, Agents, and Administrators** to perform various operations such as registration, login, complaint submission, complaint tracking, feedback, and profile management.

The frontend follows a **component-based architecture**, where the application is divided into reusable components, pages, services, and contexts. This structure improves code reusability, maintainability, and scalability while ensuring smooth communication with the backend through REST APIs.

1. public/

Development

The **public** folder stores static files that are directly served to the browser without processing by React.

Files

index.html

- Main HTML page loaded by the browser.
- Contains the root `<div>` where the React application is rendered.

favicon.ico

- Displays the application icon in the browser tab.

manifest.json

- Contains Progressive Web App (PWA) configuration.
 - Defines application name, icons, theme color, and display mode.
-

2. src/

The **src** folder contains the complete React application source code.

assets/

Development

Stores all static resources used in the frontend.

images/

Contains images used across the application.

logo.png

Application logo.

home.jpg

Home page banner image.

login.png

Login page illustration.

register.png

Registration page illustration.

admin.png

Administrator dashboard icon/image.

agent.png

Agent dashboard icon/image.

user.png

User dashboard icon/image.

avatar.png

Default profile image.

CSS/

Development

Contains all CSS files used to style the application.

Stylesheets

- **index.css** – Global styling.
- **App.css** – Root application styling.
- **Navbar.css** – Navigation bar styling.
- **Sidebar.css** – Sidebar layout styling.
- **Footer.css** – Footer styling.
- **Home.css** – Home page styling.
- **Login.css** – Login page styling.
- **Register.css** – Registration page styling.
- **UserDashboard.css** – User dashboard design.
- **AgentDashboard.css** – Agent dashboard design.
- **AdminDashboard.css** – Admin dashboard design.
- **ComplaintForm.css** – Complaint submission form styling.
- **ComplaintDetails.css** – Complaint details page styling.
- **ComplaintCard.css** – Complaint card layout.

- **Feedback.css** – Feedback page styling.
 - **Profile.css** – User profile page styling.
 - **Notification.css** – Notification panel styling.
 - **Table.css** – Table formatting.
 - **Buttons.css** – Button styles.
 - **Forms.css** – Common form styles.
 - **Loader.css** – Loading animation styling.
 - **Responsive.css** – Responsive design for mobile, tablet, and desktop.
-

components/

Development

The **components** folder contains reusable UI components used across multiple pages.

Navbar.jsx

Displays the top navigation menu.

Functions

- Navigation links
 - User profile access
 - Logout option
-

Sidebar.jsx

Displays dashboard navigation.

Functions

- Dashboard menu
 - Complaint menu
 - Reports
 - Settings
-

Footer.jsx

Displays footer information.

Loader.jsx

Displays loading animation while data is being fetched from the server.

ComplaintCard.jsx

Displays complaint summary.

Includes

- Complaint ID
 - Status
 - Department
 - Date
-

ComplaintForm.jsx

Contains complaint registration form.

Allows users to

- Select department
 - Enter complaint details
 - Upload images
 - Submit complaint
-

Notification.jsx

Displays notifications received from the server.

Examples

- Complaint Submitted
 - Complaint Assigned
 - Complaint Resolved
-

ProtectedRoute.jsx

Restricts unauthorized users from accessing protected pages.

Checks

- Login status
 - User role
 - Authentication token
-

pages/

Development

The **pages** folder contains complete pages rendered through React Router.

Home.jsx

Displays the home page.

Features

- Welcome banner
 - About system
 - Navigation
-

Login.jsx

Provides login interface.

Features

- Email
 - Password
 - Authentication
-

Register.jsx

Allows new users to create an account.

UserDashboard.jsx

Displays

- User profile
 - Complaint history
 - Complaint status
 - Notifications
-

AgentDashboard.jsx

Displays

- Assigned complaints
 - Update complaint status
 - Resolve complaints
-

AdminDashboard.jsx

Displays

- Total complaints
- Users
- Agents
- Reports

- Analytics
-

ComplaintDetails.jsx

Displays complete complaint information.

Feedback.jsx

Allows users to provide ratings and comments after complaint resolution.

Profile.jsx

Displays and updates user profile.

NotFound.jsx

Displays a custom **404 Page Not Found** screen for invalid routes.

context/

Development

The **Context API** manages shared application state without prop drilling.

AuthContext.jsx

Manages

- User Login

- Logout
 - Authentication
 - JWT Token
 - User Details
-

ComplaintContext.jsx

Manages

- Complaint List
 - Complaint Status
 - Complaint Updates
 - Shared Complaint Data
-

services/

Development

The **services** folder manages communication with the backend using Axios.

api.js

Creates a centralized Axios instance.

Stores

- Base URL
 - Headers
 - Authentication Token
-

authService.js

Handles authentication APIs.

Functions

- Register
 - Login
 - Logout
 - Forgot Password
-

complaintService.js

Handles complaint APIs.

Functions

- Register Complaint
 - Update Complaint
 - View Complaint
 - Delete Complaint
-

feedbackService.js

Handles feedback APIs.

Functions

- Submit Feedback
 - View Feedback
-

notificationService.js

Handles notification APIs.

Functions

- Get Notifications
 - Mark as Read
-

App.jsx

Development

Acts as the root component of the application.

Responsibilities

- Loads application layout
 - Configures React Router
 - Renders all pages
 - Manages common components
-

main.jsx

Development

Entry point of the React application.

Responsibilities

- Creates React root
 - Loads App component
 - Imports global CSS
 - Wraps Context Providers
-

index.css

Contains global styling applied throughout the application.

package.json

Contains

- Project metadata
 - Dependencies
 - React scripts
 - Build commands
 - Development commands
-

package-lock.json

Locks dependency versions to ensure consistent installations across development environments.